

GIULIANO PENNACCHIO

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PROFESSIONAL PROFILE

High-performing Aerospace Engineer (MSc) specialized in Flight Dynamics, GNC, and AOCS. Expertise in high-fidelity orbit propagation, state estimation (EKF, Batch Least Squares), and Phase 0/A/B1 space mission studies. Proficient in GMAT, MATLAB/Simulink (Simscape), and Python (NumPy, SciPy, Astropy, Poliastro). Proven track record in bridging orbital mechanics theory with software implementation via CubeSat mission design and space software tools.

TECHNICAL SKILLS & TOOLKIT

- GNC & Flight Dynamics:** Orbit Propagation, Orbit Determination (OD), Trajectory Optimization, Covariance Analysis, EKF/UKF Tuning, Multi-Sensor Data Fusion, Pointing Budget Analysis.
- Systems Engineering:** Early-Phase Mission Design (Phase 0/A/B1), System Trade-Offs, Mission Analysis, Payload-Launcher Interface Requirements, Requirements Management.
- Software & Simulation:** Python (NumPy, SciPy, Matplotlib, Poliastro, Astropy, FastF1, Pandas), MATLAB & Simulink (Model-Based Design, Simscape), GMAT (Ephemeris & Maneuver Modeling), STK familiarity.
- Dev & Tools:** Technical Documentation (SoW, SRD, ICD), Git, Linux/Bash Scripting, Unit Testing (pytest).

SPACE ENGINEERING PROJECTS & EXPERIENCE

IGNIS CubeSat Project | Mission Analysis & AOCS Team Member Nov 2025 - Mar 2026

- Designed and validated a full-state Extended Kalman Filter (EKF) in Simulink for AOCS state estimation and sensor fusion.
- Derived and allocated pointing budget requirements to GNC algorithms, sensors, and actuators for Phase 0/A compliance.
- Managed Interface Control Documents (ICD) across Power, OBDH, and Structure subsystems.

Independent Flight Dynamics & Space Systems Portfolio | Lead Developer Ongoing

- Interplanetary Trajectory Optimization:** Developed Python framework solving Lambert's problem for Earth-Mars transfers ($C_3=9.82 \text{ km}^2/\text{s}^2$, TOF=293.4 days); validated against NASA Mars 2020 telemetry.
- High-Fidelity Orbit Propagator:** Built numerical propagator in Python cross-validated against GMAT; modeled J2 LEO drift (~300 km over 30 days), secular drag decay (-1.8 m/day), and SSO RAAN precession.
- Walker Delta Coverage Tool:** Evaluated gap metrics across 200+ Mediterranean targets for J2-perturbed Walker Delta constellations (Mean Revisit Time: 17.52 min, Max Gap: 178min).
- Multi-Constellation Orbit Visualizer:** Developed SGP4/SDP4 Python propagation engine across LEO, MEO, GEO, and IGSO regimes. Implemented TEME-to-ITRS transformations via GMST/GAST for ground track kinematics, generated 2D maps and speed profiles.
- F1 Telemetry & Dynamics Pipeline:** Built Python toolkit (FastF1, SciPy) interpolating time-series telemetry to 1.0m spatial grid via cubic splines. Applied Savitzky-Golay filtering for spatial derivatives/curvature, velocity-squared aero loads, G-G friction circles, and ERS energy clipping detection.
- Batch Least Squares OD (WIP):** Implement Python Batch Least Squares estimator for 3-day LEO SSO tracking arcs using simulated Kiruna and Fucino ground station telemetry.
- Real-Time Orbit Determination & EKF Framework:** Developed a high-fidelity MATLAB/Simulink Extended Kalman Filter (EKF) integrating high-order gravity/perturbation dynamics and GNSS data assimilation. Achieved sub-second initial convergence and steady-state position errors under 10–20m (3σ), implementing real-time FDIR via Mahalanobis distance gating (χ^2 test) and NIS consistency testing.
- MSc Thesis - Non-Cooperative Target Pose Estimation:** Evaluated LiDAR-based OBB-TM algorithm performance for uncooperative spacecraft pose acquisition in lost-in-space conditions. Achieved a 93-95% global success rate across variable point-cloud noise levels, analyzing symmetry resolution and convergence robustness to support autonomous rendezvous and servicing GNC architectures.

EDUCATION

University of Naples Federico II | Naples, Italy Graduated: Feb 2026

MSc in Aerospace Engineering - Space Specialization | Final Grade: 106/110

- Master's Thesis:** "Performance Analysis of LIDAR-based Pose Estimation Algorithms for Non-Cooperative Space Targets."
- Core Focus:** Space GNC, GNSS Navigation, Flight Dynamics, Orbital Mechanics, Space Robotics

University of Naples Federico II | Naples, Italy Graduated: Jun 2022

BSc in Aerospace Engineering

- Bachelor's Thesis:** "Aerodynamic stall of GA(W)-1 profile via CFD."

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Space Days Professional Certificate | GNSS Academy** (May 2026) PNT architectures and GNSS technology.
- Introduction to Physical Modeling using Simscape | Skill-Lync** (Nov 2025) GNC logic and multidisciplinary simulation in Simulink.
- AI Fluency: Framework & Foundation | Anthropic** (Jul 2026) - Technical computing acceleration and data analysis.

- **Linux Unhatched | Cisco Networking Academy** (Dec 2025) - Fundamentals of Linux OS, Bash command-line interface, system navigation, and shell scripting.

LANGUAGES & CORE COMPETENCIES

- **Languages:** Italian (Native), English (Full Professional Proficiency C2 Proficient (Score: 71/100, EF SET Certificate, 2026)).
- **Core Competencies:** Systems Engineering Mindset, Technical Problem Solving, Cross-functional Communication, Multicultural Team Collaboration, Spacecraft Systems Integration.